

Norway

SLEIPNER

Volve F

F-1

F-1 C

Plan: F-1 C InjectorProducer opt3 revp2

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-1	North Reference:	Grid
Wellbore:	F-1 C	Survey Calculation Method:	Minimum Curvature
Design:	F-1 C InjectorProducer opt3 revp2	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-1					
Well Position	+N/-S	3.17 m	Northing:	6,478,566.69 m	Latitude:	58° 26' 29.9072 N
	+E/-W	-3.53 m	Easting:	435,046.49 m	Longitude:	1° 53' 14.7075 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-1 C				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	31.10.2013	-1.09	71.62	50,525

Design	F-1 C InjectorProducer opt3 revp2				
Audit Notes:					
Version:	Phase:	PLAN	Tie On Depth:	1,335.60	
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	3.17	-3.53	27.99	

Survey Tool Program	Date	14.11.2014			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
150.15	210.70	15/9 F1 36" MWD (F-1)	Magnetic, old	Magnetic Tools (MWD, EMS)	
237.60	358.50	15/9 F1 26" Gyro Scientific Modular (F-1)	NOT_useMWD gyro, Baker Hughes'	Baker Hughes' MWD gyro service (GyroTrak)	
378.00	737.40	15/9 F-1 26" SDI Keeper Gyro Single Shot	Gyro, old	Other Gyro Tools	
756.80	1,335.60	15/9 F-1 26" OnTrak (F-1)	Magnetic, IFR, mag-corr	Magnetic Tools (MWD, EMS)	
1,335.60	2,506.00	F-1 C InjectorProducer opt3 revp2 (F-1 C)	Magnetic, IFR, mag-corr, red	Magnetic Tools (MWD, EMS) without gyro-verification	
2,506.00	3,078.00	F-1 C InjectorProducer opt3 revp2 (F-1 C)	Magnetic, IFR, mag-corr, red	Magnetic Tools (MWD, EMS) without gyro-verification	
3,078.00	4,038.98	F-1 C InjectorProducer opt3 revp2 (F-1 C)	Magnetic, IFR, mag-corr, red	Magnetic Tools (MWD, EMS) without gyro-verification	

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
1,335.60	10.17	309.34	1,323.76	99.53	-108.75	35.73	0.000	0.000	0.000	
1,348.40	9.45	313.41	1,336.38	100.97	-110.39	36.23	2.342	-1.688	9.539	
1,358.40	9.45	313.41	1,346.24	102.10	-111.58	36.67	0.000	0.000	0.000	
1,378.40	8.16	303.40	1,366.01	104.01	-113.96	37.24	3.000	-1.938	-15.022	
1,478.40	17.07	281.04	1,463.55	110.74	-134.33	33.62	3.000	2.672	-6.706	

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Wellbore:	F-1 C	Survey Calculation Method:	Minimum Curvature
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Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
1,676.78	21.66	231.42	1,651.86	93.37	-191.92	-8.74	2.500	0.694	-7.503	
2,111.51	21.66	231.42	2,055.91	-6.66	-317.34	-155.92	0.000	0.000	0.000	
3,265.32	78.67	41.28	2,942.90	392.61	-35.03	329.13	2.600	1.482	4.416	
3,631.85	78.67	41.28	3,014.90	662.71	202.05	678.89	0.000	0.000	0.000	
3,938.98	48.80	56.41	3,149.90	844.77	403.12	934.02	3.193	-2.917	1.478	
4,038.98	48.80	56.41	3,215.77	886.40	465.80	1,000.19	0.000	0.000	0.000	

Design Targets										
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude		Longitude
F-1 C Target#1 - plan hits target center - Point	0.00	360.00	2,942.90	392.61	-35.03	6,478,956.00	435,015.00	58° 26' 42.4752 N		1° 53' 12.3688 E
F-1 C Target#2 - plan hits target center - Point	0.00	360.00	3,014.90	662.71	202.05	6,479,226.00	435,252.00	58° 26' 51.3297 N		1° 53' 26.7068 E
F-1 C Target#3 - plan hits target center - Point	0.00	0.00	3,149.90	844.77	403.12	6,479,408.00	435,453.00	58° 26' 57.3201 N		1° 53' 38.9161 E

Casing Points						
Measured Depth (m)	Vertical Depth (m)	Name			Casing Diameter (in)	Hole Diameter (in)
220.90	220.90	30"			30.000	36.000
1,348.40	1,336.38	20"			20.000	26.000
2,506.00	2,442.69	13 3/8"			13.375	17.500
3,078.00	2,880.80	9 5/8"			9.625	12.250
4,038.98	3,215.76	7"			7.000	8.500

Checked By: _____	Approved By: _____	Date: _____
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